Juan Barrera
Professor Mark Voortman
Advanced Topics in AI
Final Project

Campus AI Final Project

Integrated Multi-Agent Simulation, Neural NLP, EM Hidden-Mode Inference, and Q-Learning

1. Domain and Problem Context

This project models a university campus as an intelligent multi-agent environment. The domain contains student agents moving through campus locations such as Academic Hall, West Penn, Student Center, Library, Village Park, Track, Playhouse, George Rowland White, residence halls, and Off Campus. The goal is not only to display movement on a map, but to create agents that can reason about where they should be, explain why they are there, interact socially, respond to natural-language questions, and make autonomous decisions with consequences.

The simulation includes 50 agents representing athletes, COPA/performance students, and regular students. Each agent has a name, type, personality, housing status, expected location, actual location, next destination, missed-class count, reliability score, academic risk, and rebel status. A rebel agent is an agent whose actual location differs from the location it is expected to occupy during the current time slot.

Main problem being solved

- How can an AI system represent the expected and actual locations of many student agents over time?
- How can the system explain why an agent is following or deviating from its schedule?
- How can a user communicate with agents in natural language and receive grounded responses?
- How can agents decide whether to follow commands, miss class, attend practice, or rebel?
- How can the system expose AI evidence, such as attention weights, EM mode probabilities, and Q-values?

Actual implementation claim

The project contains actual implemented AI components: a neural NLP intent classifier with embeddings and attention, supervised classification, expected-utility reasoning, value of information, multi-agent congestion analysis, EM hidden activity-mode inference, and Q-learning reinforcement learning. Some methods are implemented directly as algorithms; others are applied reasoning layers adapted to the campus simulation.

System architecture

Layer	Role	Output
React frontend	Campus map, selected agent card, question panel, rebel list	Visual movement and user interaction
Flask backend	Receives selected agent state and question	JSON response with answer and AI evidence
Neural NLP	Classifies intent and returns attention/embeddings	predicted_intent, confidence, attention
Campus reasoner	Utility, VOI, congestion, probability, Q-style reasoning	explainable campus evidence
EM + Q-learning	Hidden activity mode and learned policy recommendation	mode probabilities and learned Q-values

2. Inputs, Outputs, and Agent State

The system combines time, agent profiles, schedule expectations, actual behavior, user questions, and the multi-agent environment. These inputs allow the AI system to answer not only where an agent is, but why it is there, whether it is expected to be there, who else is nearby, and what decision it should make next.

Inputs

- Simulation time: 00:00-08:00 sleep, morning/class slots, 15:00-17:30 practice, 18:00-21:00 evening, and 21:00-00:00 free time.
- Agent profile: student type, sport, housing, personality, reliability score, academic risk, missed classes, and current conversation state.
- Expected schedule: where the agent should be during the current time frame.
- Actual behavior: where the agent currently is, including rebel deviations.
- User text: natural-language questions or commands such as 'Who are you with?', 'Why are you here?', 'Can we miss class?', or 'Please go to Village Park next.'
- Multi-agent context: location of all other agents, crowding, companions, and destination conflicts.

Outputs

- Map movement along campus paths rather than straight-line teleporting.
- Selected-agent profile with expected location, actual location, next destination, rebel status, missed classes, reliability, and academic risk.
- Rebel Agents list showing agents whose actual location differs from expected location.
- Natural-language response generated from the selected agent state and predicted intent.
- AI evidence including predicted intent, confidence, attention weights, word representations, EM hidden-mode probabilities, Q-learning recommendation, and campus reasoning evidence.

Expected vs. actual location

A major design decision was to separate expected location from actual location. The expected location is the location assigned by the schedule generator. The actual location is where the agent chooses to be. This allows the system to represent autonomy. For example, an agent may be expected at Academic Hall but actually be at Student Center. In that case, the agent becomes a rebel agent and can explain why it chose a different location.

Field	Meaning
expected_location_now	Where the agent is supposed to be at the current time
current_location	Where the agent actually is
is_rebel	True when actual location differs from expected location
rebel_reason	Personality/autonomy reason for deviation
conversation_state	Tracks whether the agent is waiting for a user justification
missed_classes, reliability_score, academic_risk	Memory and consequences after skipping obligations

Conversation and decision flow

When an agent is asked to miss class or leave an obligation, it does not always make a one-turn decision. If the request requires justification, the backend returns a conversation state such as awaiting_miss_class_reason or awaiting_command_reason. The next user message is interpreted as the reason, and the agent accepts or refuses based on personality, obligation type, reason quality, academic risk, and learned reinforcement-learning evidence.

3. AI Methods by Book Chapter: Chapters 15-18

The final system integrates several chapters from the AI course text. The methods are not isolated scripts anymore; they are connected in one campus simulation where a user can inspect the output of each AI layer.

Chapter 15 - Probabilistic Reasoning over Time

The simulation uses time slots as temporal states. Each agent's expected and actual location changes as the simulated day progresses. Because agents can deviate from schedule, the system represents uncertainty about student behavior. This chapter supports the location-state model, temporal transitions, and hidden uncertainty in agent movement.

Chapter 16 - Making Simple Decisions

The campus reasoner applies expected utility to compare possible destinations. Utility is based on schedule fit, distance, crowding, personality preferences, and whether the location satisfies the current obligation. Value of information is represented by whether checking a destination or crowd level would improve the movement decision.

Chapter 17 - Making Complex Decisions

Sequential decision-making appears in the time-slot transition system and movement paths. The agent must reason over a sequence of states rather than a single instant. Multi-agent decision concepts appear through congestion analysis, shared locations, companion detection, and the Rebel Agents panel. The system also handles multiple agents attempting to occupy or target the same locations.

Chapter 18 - Learning from Examples

The NLP intent classifier is trained from examples of questions mapped to intent labels such as identity_query, social_presence_query, absence_reason, miss_class_request, and user_instruction. The model returns predicted intent, confidence, attention weights, and evaluation metrics. This chapter is also represented through supervised learning principles: labeled data, model evaluation, accuracy, macro-F1, and classification reports.

Chapter	Project implementation	Problem solved
15	Time slots, location uncertainty, expected vs actual location	Track agents over time
16	Expected utility and VOI	Choose/explain destinations
17	Sequential movement, multi-agent congestion	Coordinate many agents
18	Supervised NLP intent classification	Map questions to intents

Implementation status

These chapters are implemented as active system behavior. When a user asks an agent a question, the response can include the current time state, expected/actual locations, utility scoring, multi-agent congestion, and the predicted NLP intent.

4. AI Methods by Book Chapter: Chapters 20-23

Chapter 20 - Learning Probabilistic Models with EM

The EM activity model treats the true activity mode of an agent as hidden. Possible hidden modes include CLASS_MODE, STUDY_MODE, PRACTICE_MODE, SOCIAL_MODE, REST_MODE, REBEL_MODE, and FREE_TIME_MODE. The observed variables include student type, personality, time bucket, current location category, expected location category, and rebel status. The E-step estimates posterior responsibility for each hidden mode, and the M-step updates categorical probability tables. The system returns the most likely hidden mode, mode probabilities, iterations, and log-likelihood.

Chapter 21 - Reinforcement Learning with Q-Learning

The Q-learning module trains a Q-table over abstract campus states. A state includes student type, personality, time bucket, obligation status, alignment/rebel status, academic-risk level, and missed-class level. Actions include FOLLOW_EXPECTED, STAY_CURRENT, GO_SOCIAL, GO_HOME, GO_LIBRARY, and REBEL. Rewards encourage attending class/practice obligations, matching personality preferences during free time, and avoiding repeated academic risk. The backend returns learned Q-values and a recommended action.

Chapters 22-23 - Natural Language Processing and Communication

The user communicates with agents using text. The system classifies the intent, extracts locations from questions, checks politeness and reason quality, handles social queries, and manages multi-turn conversations. If a user asks an agent to leave an obligation, the system activates a conversation state and requires a stronger reason before deciding.

Chapters 22-23 - Neural NLP, Attention, and Transfer Learning

The neural NLP system is connected to the book chapters on natural language processing and communication. It implements word representations, GRU sequence modeling, Attention-BiGRU classification, attention weights, nearest-neighbor embedding evidence, model metrics, and an optional DistilBERT transfer-learning extension. These features support language understanding and agent communication in the final system.

Method	Actual implementation evidence
EM	E-step, M-step, hidden modes, posterior probabilities, log-likelihood
Q-learning	Q-table, actions, reward function, training episodes, learned Q-values
Neural NLP	Embeddings, GRU/BiGRU, attention weights, confidence, word neighbors
Communication	Intent classification, command handling, follow-up states, consequences

Direct vs. applied implementations

EM, Q-learning, supervised neural NLP, attention, and word representations are implemented directly as algorithms. Expected utility, value of information, congestion reasoning, and sequential movement are implemented as applied reasoning layers inside the campus simulation.

5. System Implementation, Results, Limitations, and Conclusion

Implementation

The frontend is a React application that displays the campus map, agent dots, selected-agent panel, question panel, AI system status, campus legend, and Rebel Agents list. The backend is a Flask API that receives the selected agent and question, calls the neural NLP engine, campus AI reasoner, EM model, and Q-learning model, then returns one integrated JSON response.

Results

- The system successfully separates expected location from actual location and identifies rebel agents.
- Users can click any agent or rebel-agent card and ask grounded questions.
- Agents can explain schedule compliance, deviation, social companions, free-time behavior, and obligations.
- The system uses conversation state for miss-class requests and obligation-conflicting commands.
- The UI displays AI evidence: intent classification, attention weights, word representations, EM mode probabilities, Q-values, expected utility, VOI, congestion, and probability estimates.

Limitations

- The environment is a simulation rather than real campus sensor data.
- Some probability distributions are synthetic and should be replaced with real observations in future work.
- The Q-learning state space is abstracted, so it learns high-level policy behavior rather than exact physical paths.
- Transfer learning is available as an extension, but the deployed system should clearly state whether DistilBERT is active or optional.

Conclusion

The project solves the core problem of building an AI-driven, explainable, interactive campus simulation. It combines temporal reasoning, rational decision-making, sequential movement, multi-agent awareness, supervised NLP, EM hidden-variable inference, Q-learning, and neural attention-based language understanding. The result is an integrated AI system where agents do not merely move; they explain, decide, remember consequences, and respond to users in natural language.

Chapter contribution summary

Chapter	How it helped solve the problem
15	Represents time and uncertain location states
16	Supports utility, preferences, and VOI
17	Supports sequential decisions and multi-agent reasoning
18	Provides supervised intent classification and evaluation
20	Uses EM to infer hidden activity modes
21	Uses Q-learning to recommend actions from rewards
22-23	Enables language-based communication and semantic interpretation
	Implements neural embeddings, GRU/BiGRU, attention, and transfer extension

Overall, the project is not only conceptually aligned with the book; it implements selected book methods as working system components and exposes their evidence during user interaction.

Chapters Not Used Directly: Chapters 24–27

Chapter 24 Perception

Chapter 24 focuses on perception, especially computer vision topics such as image formation, image-processing operations, object recognition, and reconstructing three-dimensional worlds. These methods are important for AI systems that receive visual input from cameras or image sensors.

This project does not use Chapter 24 directly because the campus map is not interpreted through computer vision. The map is used as a visual interface, but the system does not analyze images, detect objects, classify buildings from pixels, or reconstruct a 3D environment. The agent locations are generated from structured simulation data rather than perceived from an image.

For that reason, Chapter 24 was not a direct fit for this project. If the project were expanded in the future, Chapter 24 could be used by adding a camera-based or image-based perception system that identifies campus buildings, detects student movement from video, or recognizes crowded areas from visual input.

Chapter 25 Robotics

Chapter 25 focuses on robotics, including robot hardware, robotic perception, planning to move, uncertain movement, robotic software architectures, and application domains. These topics are designed for systems that control physical robots or simulated robots with sensors and actuators.

This project does not use Chapter 25 directly because the agents are not physical robots. They do not have motors, sensors, robotic arms, wheels, or real-world movement constraints. The movement in the project is a simulated campus-path animation, not robotic motion planning with actuators or uncertain physical control.

However, the project is loosely related to robotics because the agents move through a mapped environment and make decisions about where to go. Still, it would be inaccurate to claim that the project implements robotics. A future version could use Chapter 25 by turning the campus agent into a robotic guide that physically navigates campus paths and interacts with students.

Chapter 26 Philosophical Foundations

Chapter 26 discusses philosophical foundations of AI, including weak AI, strong AI, machine consciousness, ethics, and the risks of developing artificial intelligence. These topics are important for evaluating what AI systems mean and how they should be used responsibly.

This project does not use Chapter 26 as an implementation chapter because it is not building a philosophical argument or testing whether machines can truly think. The project focuses on practical AI methods: probabilistic reasoning, utility-based decisions, reinforcement learning, EM inference, and neural NLP.

Even though Chapter 26 was not implemented as code, it is still relevant to the discussion section. The project raises ethical questions about simulated student behavior, decision-making, autonomy, and how AI systems should explain their reasoning. For example, the system models “rebel agents” and decisions about missing class, so it is important to clarify that these are simulations and not recommendations for real student behavior.

Chapter 27 AI: The Present and Future

Chapter 27 focuses on the present and future of AI, including agent components, agent architectures, current directions, and what AI does successfully. This chapter is broader and more reflective than technical.

This project does not use Chapter 27 as a direct algorithmic implementation because the chapter is not centered on one specific method like EM, Q-learning, or neural networks. Instead, it provides a high-level view of how AI systems are organized and where AI is heading.

However, Chapter 27 connects to the project conceptually because the final system is an integrated AI agent architecture. It combines multiple components: simulation, reasoning, NLP, reinforcement learning, probabilistic inference, and user interaction. Therefore, Chapter 27 can be discussed in the conclusion as a way to frame the project as an example of an integrated AI system, but it should not be listed as a directly implemented algorithmic chapter.